Day-5
Input, Typecasting & Basic Calculations

What is Input?

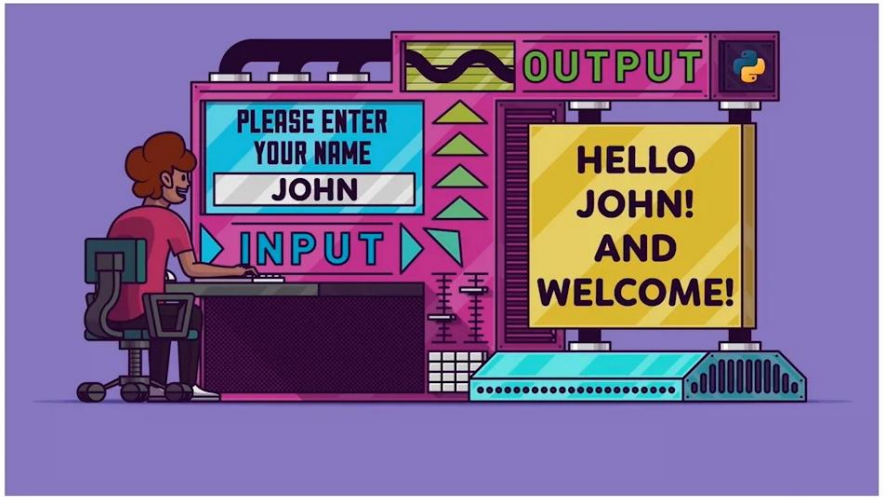
- Input means **taking information from the user**.
- Just like asking someone:
  "What is your name?"
  "What is your age?"

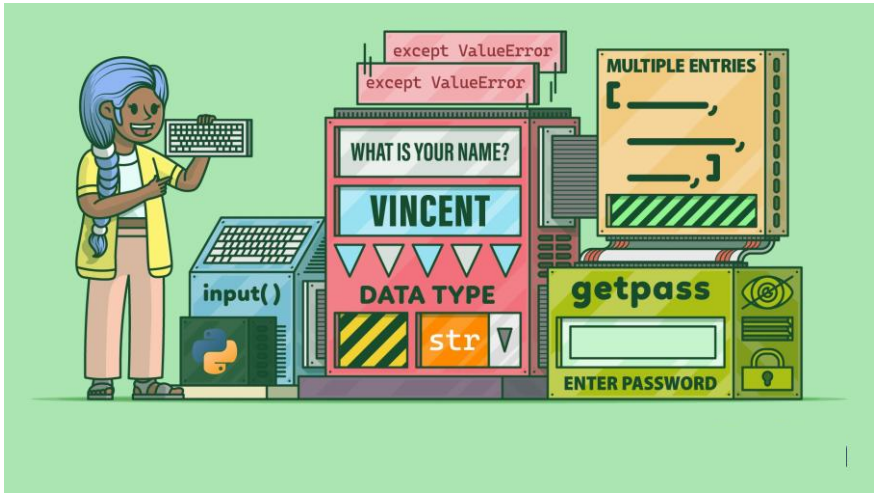
- Technical Definition
  - input() is a Python function used to **take data from the user** during program execution. Whatever the user types → **Python saves it as a string** (text) by default.

- Syntax
  - variable = input("Your message here: ")

Why Input is Important?

- Making dynamic programs
- Taking customer data
- Taking numbers from users
- Accepting filters for analysis
- Taking file path or file name
- Real-time calculators (salary, bill, score)





Problem With Input (Very Important)

Everything you type using input becomes a **string**, even numbers.

```
age = input("Enter age: ")
```

Even if the user types **25**, Python treats it like "25" → text.

This creates calculation problems.

TYPECASTING in Python

What is Typecasting?

- Typecasting means **changing the type** of the data.
- Like converting:
- text → number
- number → text
- decimal → whole number
- It's like telling Python: "Hey, this is not text. This is a number. Use it like a number."

Technical Definition

- Typecasting means converting one data type into another using built-in Python functions such as:

- int() → convert to integer
- float() → convert to decimal number
- str() → convert to text

Typecasting Examples

Converting text input to integer

- age = int(input("Enter your age: "))
- print("Age after 10 years =", age + 10)

convert text input to decimal (float)

- price = float(input("Enter product price: "))
- gst = float(input("Enter GST %: "))
- final_price = price + (price * gst / 100)
- print("Final Price =", final_price)

Convert number to string

- sales = 50000
- text = "Total sales: " + str(sales)
- print(text)

Total Sales Calculator

```
product = input("Product Name: ")
quantity = int(input("Enter quantity sold: "))
price_per_unit = float(input("Enter price per unit: "))

total_sales = quantity * price_per_unit

print("Product:", product)
print("Total Sales Amount =", total_sales)
```

Common Mistakes Students Make

❌ Forgetting to convert input → number

❌ Adding string + number → error

❌ Using int() on decimal values

❌ Not spacing messages

Assignment : Salary Slip Calculator

- **"Create a Python Program to Generate an Employee Salary Slip"**
- Write a Python program that asks the user to enter:
- Employee Name
- Basic Salary
- Bonus Amount
- Tax Percentage

- Your program should calculate:
- **Gross Salary** = Basic Salary + Bonus
- **Tax Amount** = (Gross Salary × Tax %) / 100
- **Net Salary** = Gross Salary – Tax Amount
- Finally, print a clean and clear **Salary Slip** showing all the values.

Solution Given Below

Assignment Solution

```
name = input("Employee Name: ")
basic_salary = float(input("Enter Basic Salary: "))
bonus = float(input("Enter Bonus Amount: "))
tax_percentage = float(input("Enter Tax %: "))

gross_salary = basic_salary + bonus
tax_amount = gross_salary * (tax_percentage / 100)
net_salary = gross_salary - tax_amount

print("\n--- Salary Slip ---")
print("Employee:", name)
print("Gross Salary:", gross_salary)
print("Tax:", tax_amount)
print("Net Salary:", net_salary)
```

Practice Example : - Discount Price Calculator

```
item = input("Enter Item Name: ")
price = float(input("Original Price: "))
discount = float(input("Discount Percentage: "))

discount_amount = price * discount / 100
final_price = price - discount_amount

print("Final Price of", item, "=", final_price)
```

Practical Example — Student Marks Percentage

```
student = input("Enter Student Name: ")
marks_obtained = float(input("Marks Obtained: "))
total_marks = float(input("Total Marks: "))

percentage = (marks_obtained / total_marks) * 100

print(student, "scored", percentage, "%")
```